Quadratics: Expanding & Factorising

Practice test 1 — answer sheet · 40 marks

Section A — Expanding double brackets

A1. $x^2 + 7x + 4x + 28$ (1) = $\mathbf{x^2 + 11x + 28}$ (1)

A2. $x^2 + 3x – 5x – 15$ (1) = $\mathbf{x^2 – 2x – 15}$ (1)

A3. $(x – 6)(x – 6) = x^2 – 6x – 6x + 36$ (1) = $\mathbf{x^2 – 12x + 36}$ (1)

A4. $6x^2 – 15x + 4x – 10$ (1) = $\mathbf{6x^2 – 11x – 10}$ (1)

Section B — Expanding three brackets

B1. $(x + 1)(x + 2) = x^2 + 3x + 2$ (1) $(x^2 + 3x + 2)(x + 3) = x^3 + 3x^2 + 3x^2 + 9x + 2x + 6$ (1) = $\mathbf{x^3 + 6x^2 + 11x + 6}$ (1)

B2. $(x – 2)(x + 4) = x^2 + 2x – 8$ (1) $(x^2 + 2x – 8)(2x + 1) = 2x^3 + x^2 + 4x^2 + 2x – 16x – 8$ (1) = $\mathbf{2x^3 + 5x^2 – 14x – 8}$ (1)

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Two numbers that multiply to 20 and add to 9: 4 and 5 (1) **(x + 4)(x + 5)** (1)

C2. Multiply to –28, add to –3: –7 and 4 (1) **(x – 7)(x + 4)** (1)

C3. Difference of two squares: **(x + 7)(x – 7)** (1)

C4. $2(x^2 – 25)$ (1) = **2(x + 5)(x – 5)** (1)

Section D — Factorising $ax^2 + bx + c$

D1. $2 \times 3 = 6$; numbers that multiply to 6 and add to 7: 6 and 1, so $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3)$ (1) **(2x + 1)(x + 3)** (1)

D2. $3 \times –8 = –24$; numbers: –12 and 2, so $3x^2 – 12x + 2x – 8 = 3x(x – 4) + 2(x – 4)$ (1) **(3x + 2)(x – 4)** (1)

D3. $2(2x^2 + 5x – 3)$ (1) = **2(2x – 1)(x + 3)** (1)

Section E — Using factorising

E1. $(x – 7)(x + 2) = 0$ (1) $x – 7 = 0$ or $x + 2 = 0$ (1) **x = 7 or x = –2** (1)

E2. Numerator = $(x + 2)(x + 3)$ (1) denominator = $(x + 3)(x – 3)$ (1) cancel $(x + 3)$: **(x + 2) / (x – 3)** (1)

Section F — Problem solving & proof

F1. $(n^2 + 10n + 25) – (n^2 – 10n + 25)$ (1) = $n^2 + 10n + 25 – n^2 + 10n – 25$ = **20n** (1)

F2. (a) $(x + 6)(x – 2) = x^2 + 4x – 12$ (1) $x^2 + 4x – 12 = 33$, so $x^2 + 4x – 45 = 0$ (1)
(b) $(x + 9)(x – 5) = 0$ so $x = 5$ or $x = –9$ (1) a length can't be negative, so $x = 5$ and the **width = 3 cm** (1)

F3. $(53 + 47)(53 – 47) = 100 \times 6$ = **600** (1)

Marking note: the marks in brackets show where each mark is earned. Award method marks for a correct method even if there is a later slip. Accept answers in any equivalent form, e.g. terms in a different order, or brackets written the other way round: $(x – 7)(x + 4)$ is the same as $(x + 4)(x – 7)$. “Factorise fully” questions lose the last mark if a common factor is left inside a bracket.